PERIODIC TRENDS

General Chemistry I, Lecture Series 8

Qixi Mi

Reading:

OGB8 §3.2, §§3.4–3.6, §5.5

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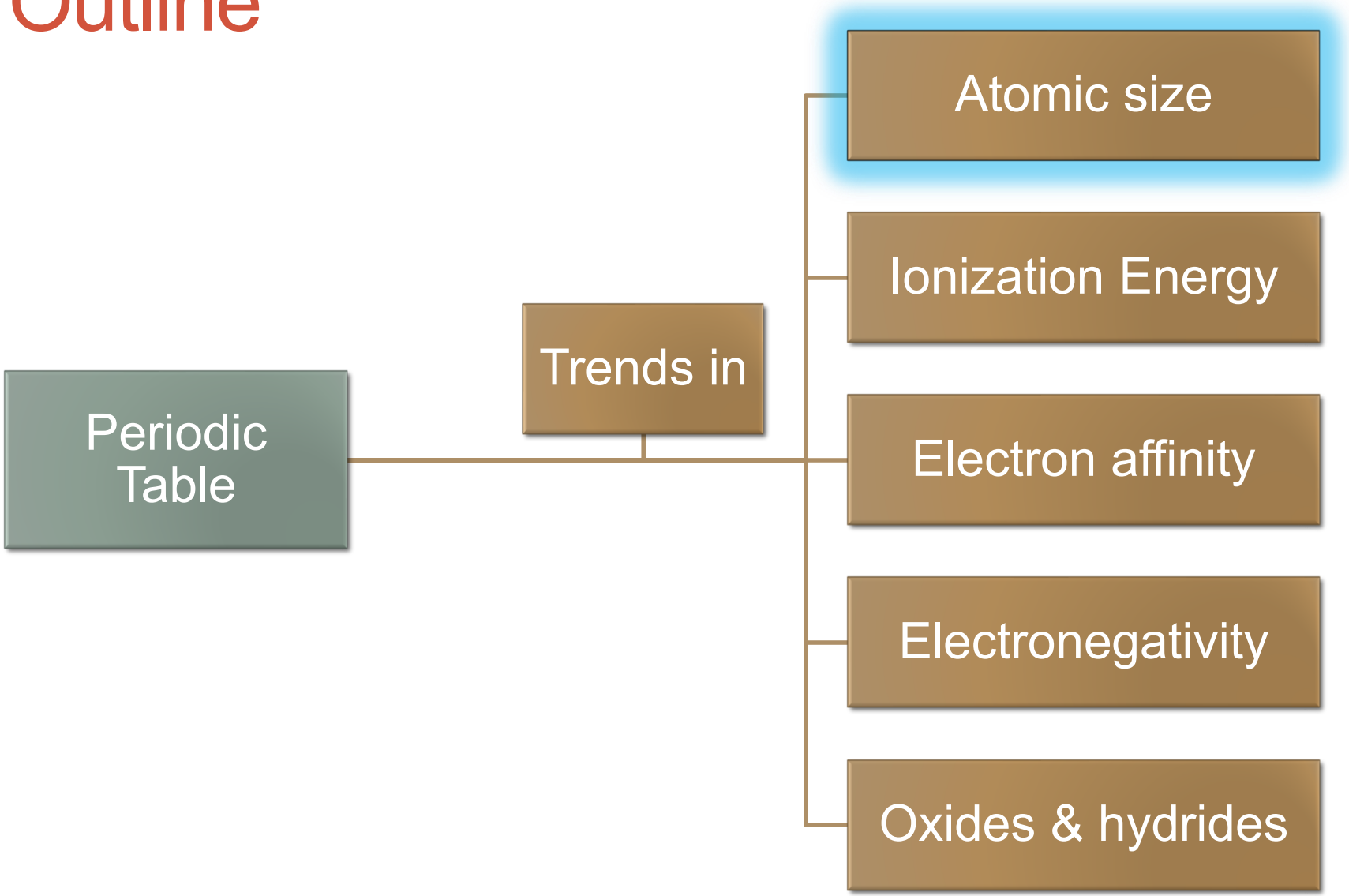
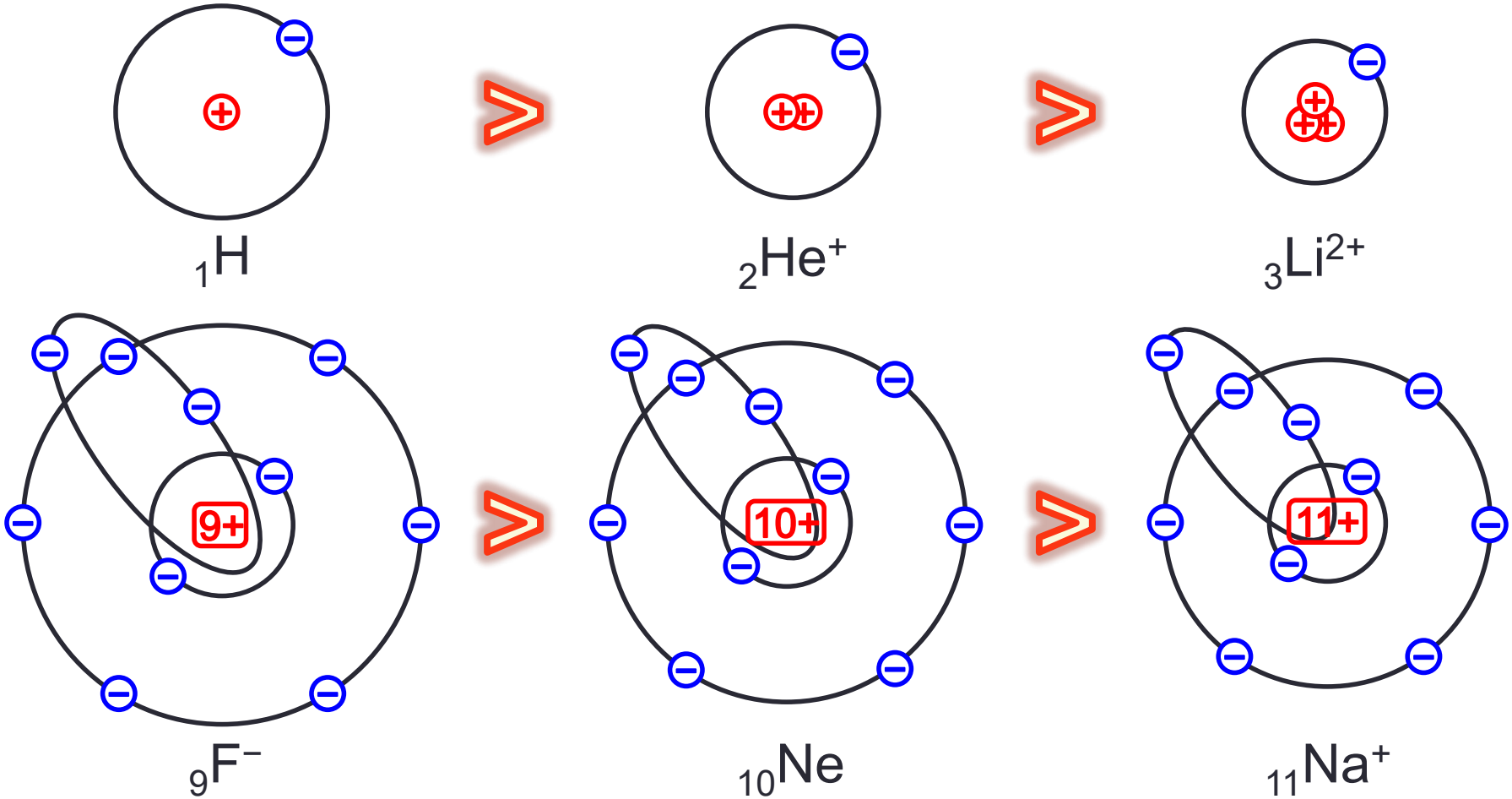
Outline


Periodic Table
Trends in
Atomic size
Ionization Energy
Electron affinity
Electronegativity
Oxides & hydrides

Radii of Isoelectronic (等电子的) Ions



Atomic/ionic radius = radius of outermost orbital

Radii of Atoms with Same *n*

Atom	Z	$Z_{eff}(2s)$	Valence shell
Li	3	~1.30	$2s^1$
Be	4	~1.95	$2s^2$
B	5	~2.60	$2s^2 2p^1$
C	6	~3.25	$2s^2 2p^2$

John C. Slater
(MIT, 1900–1976)

For *s* and *p* subshells,

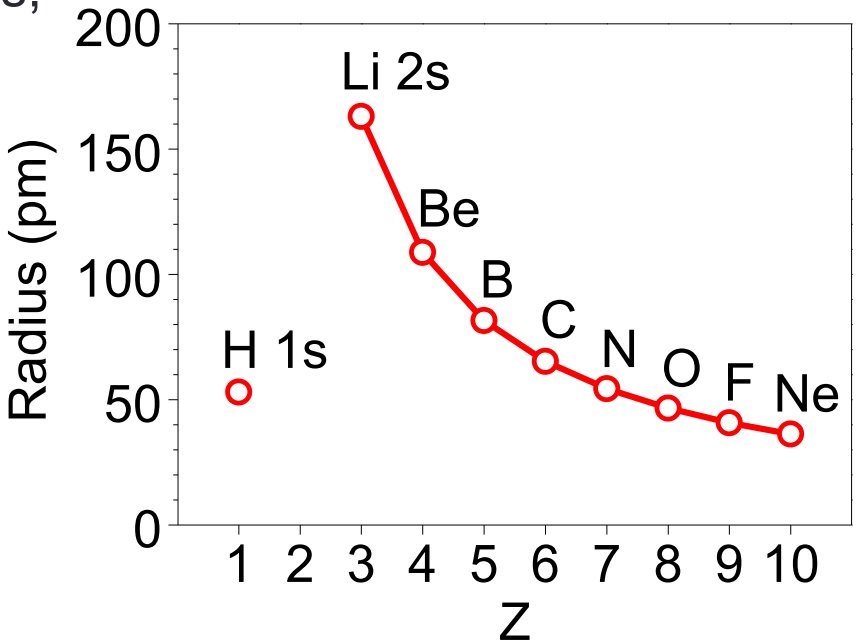
$\Delta Z_{eff}(ns) \approx 0.65\,\Delta Z$

For 2*s* and 2*p*,

$Z_{eff} \approx 0.65(Z - 1)$,

so $r(Z) = \frac{n^2}{Z_{eff}} a_0$

$\approx \frac{4a_0}{0.65} \cdot \frac{1}{Z - 1}$



Radii of Atoms with Same *n* (2)

Atom	Z	$Z_{eff}(4s)$	Valence shell
Mn	25	5.28	$3d^5 4s^2$
Fe	26	5.43	$3d^6 4s^2$
Co	27	5.58	$3d^7 4s^2$
Ni	28	5.71	$3d^8 4s^2$

Slater's Rules (1930)
OGB8 p.200

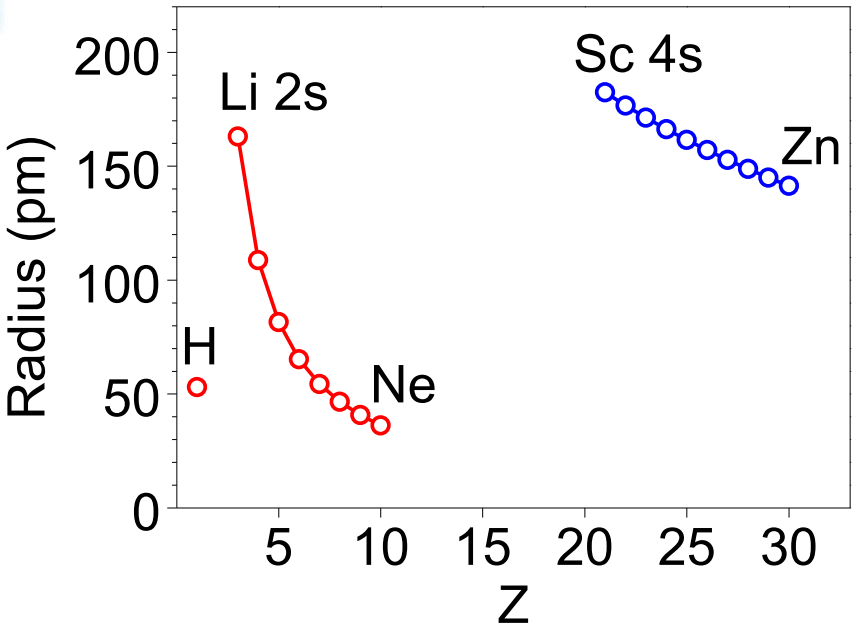
For *d* subshells, $\Delta Z_{eff}(ns) \approx 0.15\,\Delta Z$;
For *f* subshells, $\Delta Z_{eff}(ns) \approx 0$.

For 4*s*,

$Z_{eff} \approx 0.15Z + 1.53$,

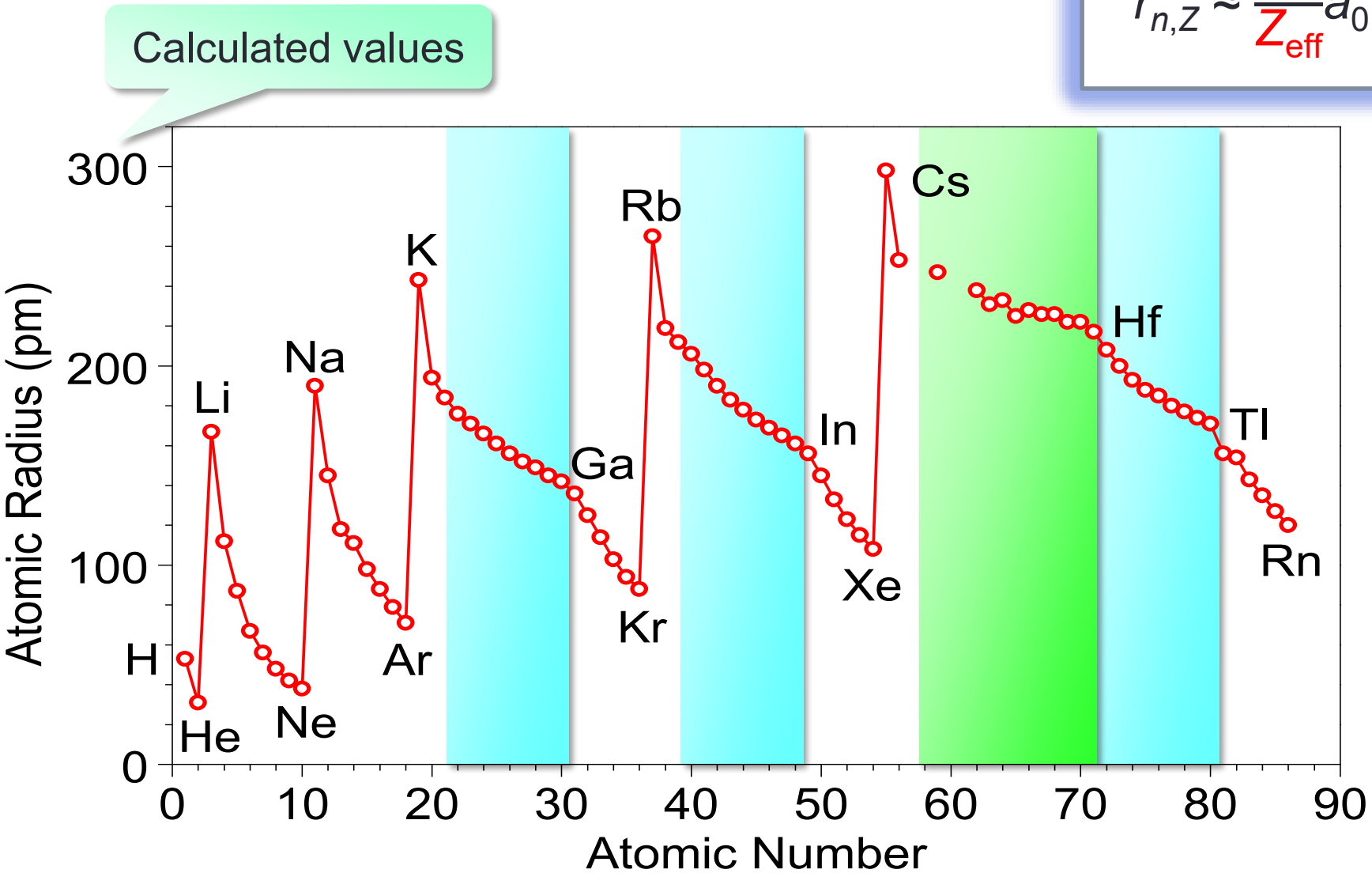
so $r(Z) = \frac{n^2}{Z_{eff}} a_0$

$\approx \frac{16a_0}{0.15} \cdot \frac{1}{Z + 10}$

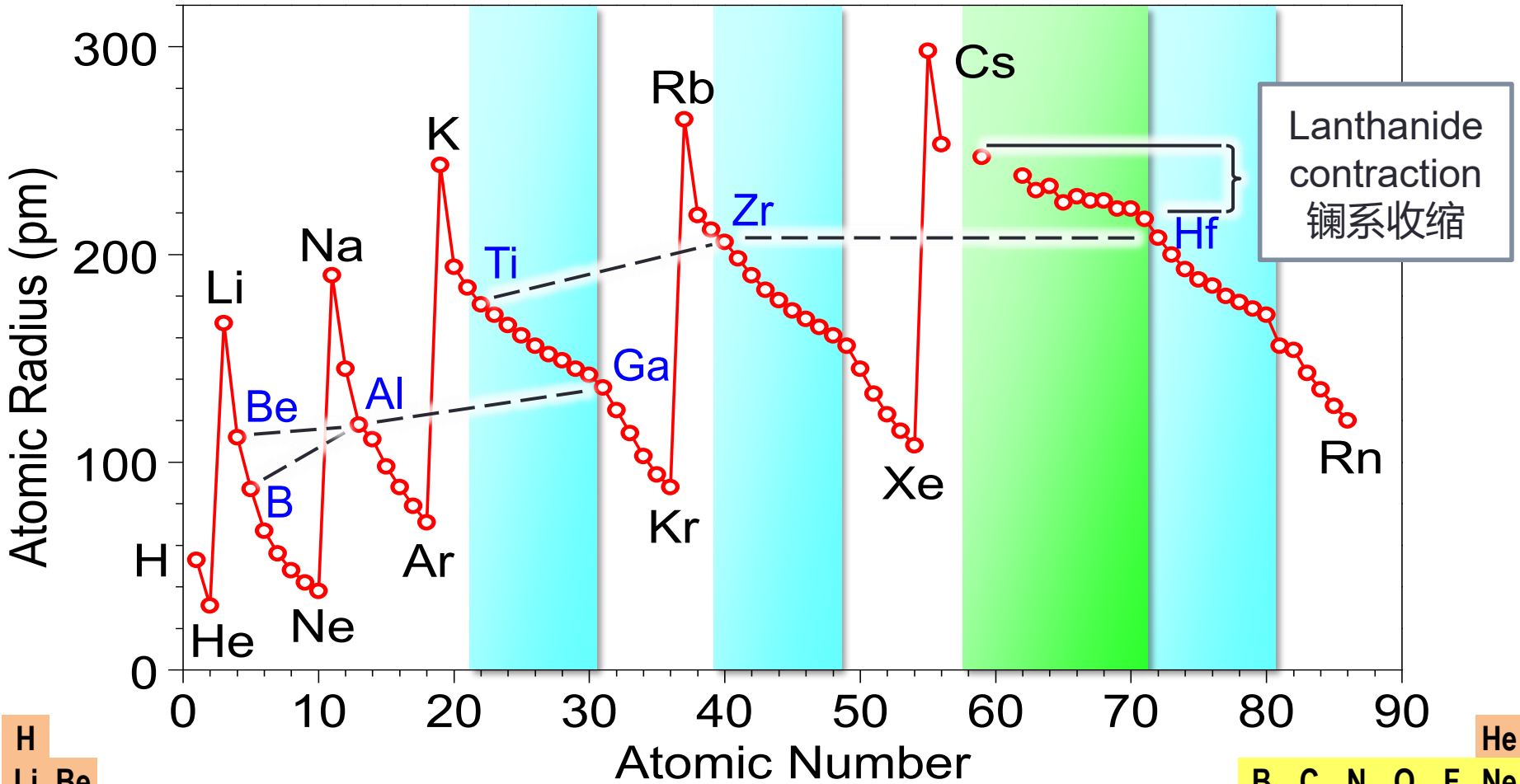


Trends in Atomic Radii (1)

$$r_{n,Z} \approx \frac{n^2}{Z_{\text{eff}}} a_0$$

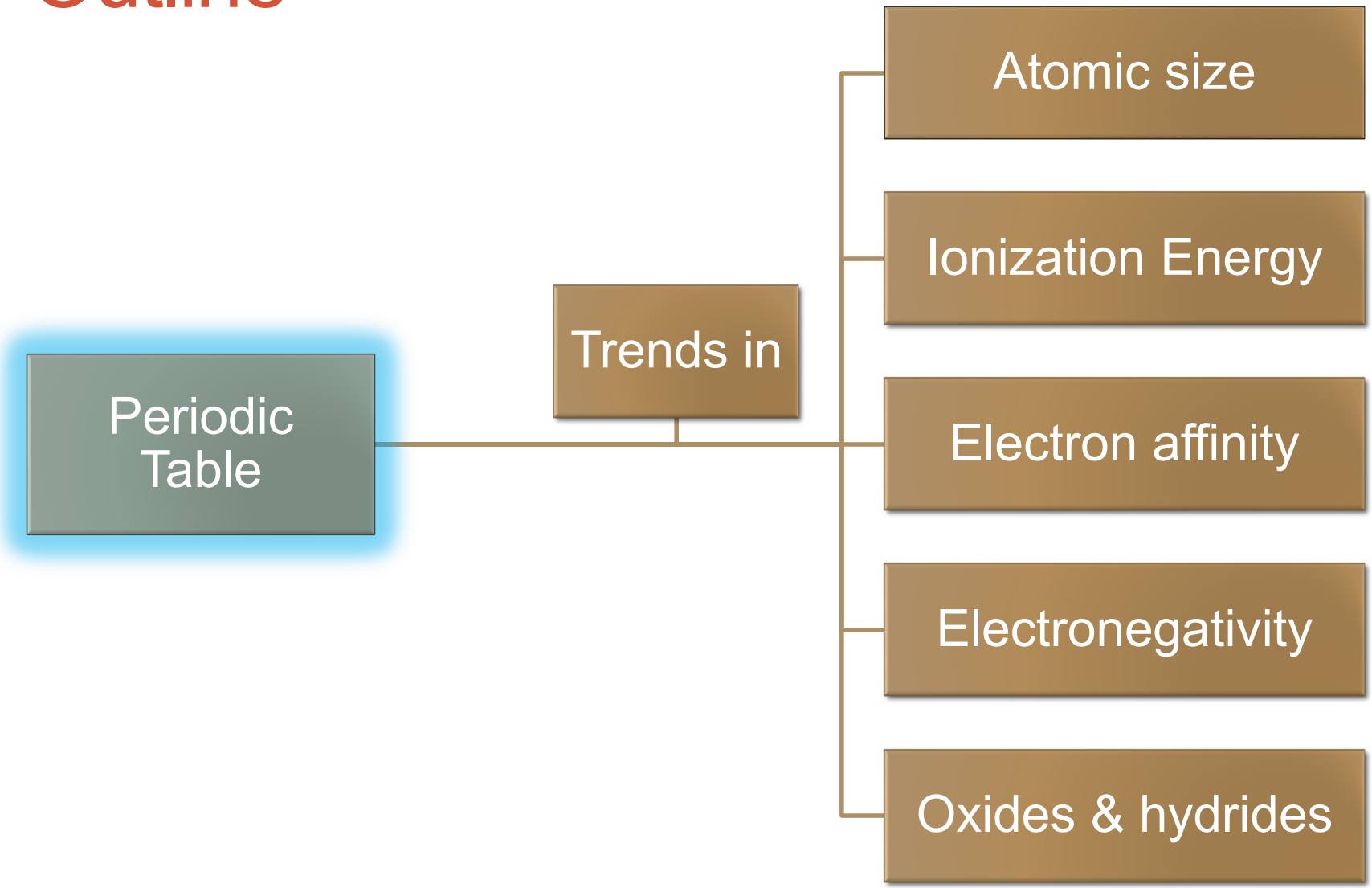


Trends in Atomic Radii (2)



1	H																		He
2	Li	Be												B	C	N	O	F	Ne
3	Na	Mg												Al	Si	P	S	Cl	Ar
4	K	Ca							Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr							Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn

Outline


Periodic Table
Trends in
Atomic size
Ionization Energy
Electron affinity
Electronegativity
Oxides & hydrides

2019 IYPT
International Year of the Periodic Table of Chemical Elements

The Periodic Table

Group 族 ⟶

IUPAC #

CAS #

Period 周期 ↓

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIIIB			IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	H																	He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	**															

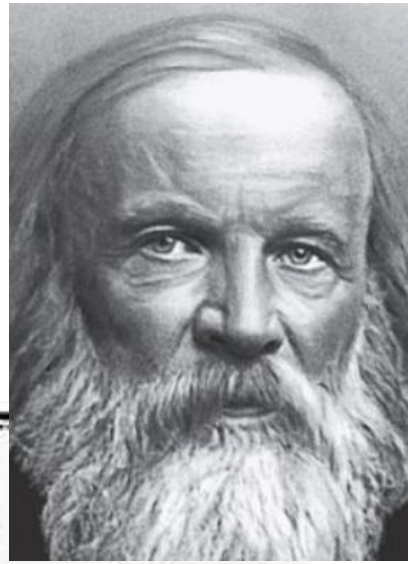
s block *p* block
d block *f* block

*	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
**	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Mendeleev's Periodic Table

(1871)

Dmitri Mendeleev
(St. Petersburg,
1834–1907)



Reihen	Gruppo I. — R^2O	Gruppo II. — RO	Gruppo III. — R^2O^3	Gruppo IV. RH^4 RO^2	Gruppo V. RH^3 R^2O^5	Gruppo VI. RH^2 RO^3	Gruppo VII. RH R^2O^7	
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,3	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Cu=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ru=104, Rh=104, Pd=106, Ag=108.
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er=178	?La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	— — — —

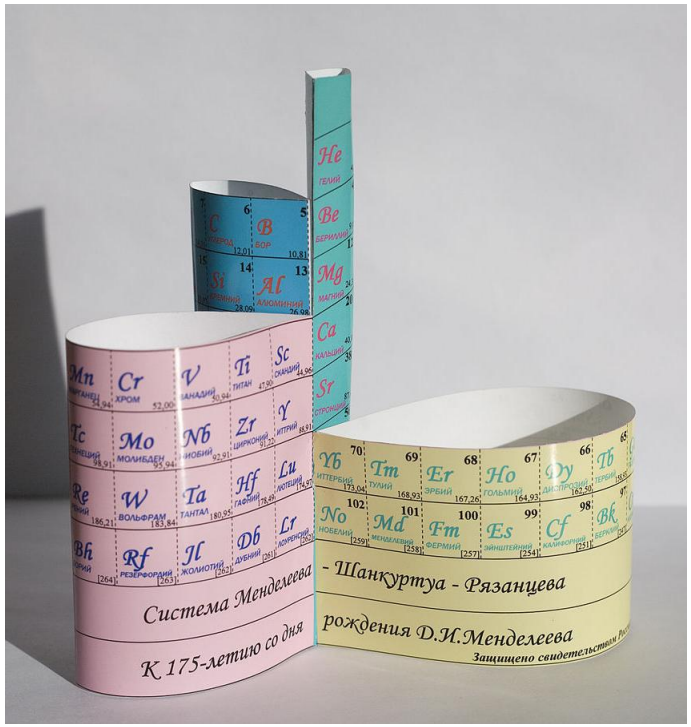
Alternative Periodic Tables

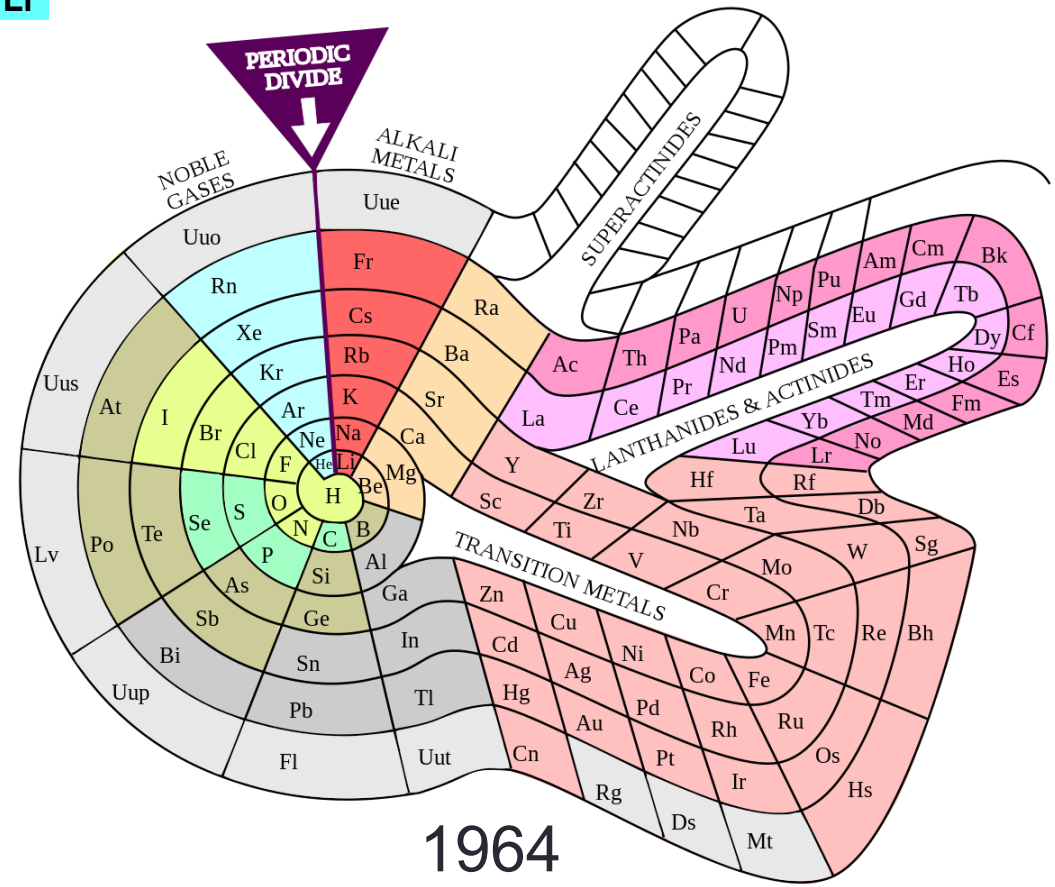
1928

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No

																H	He
																Li	Be
												B	C	N	O	F	Ne
												Al	Si	P	S	Cl	Ar
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn			Ga	Ge	As	Se	Br	Kr
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd			In	Sn	Sb	Te	I	Xe
Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg			Tl	Pb	Bi	Po	At	Rn
Lr																	

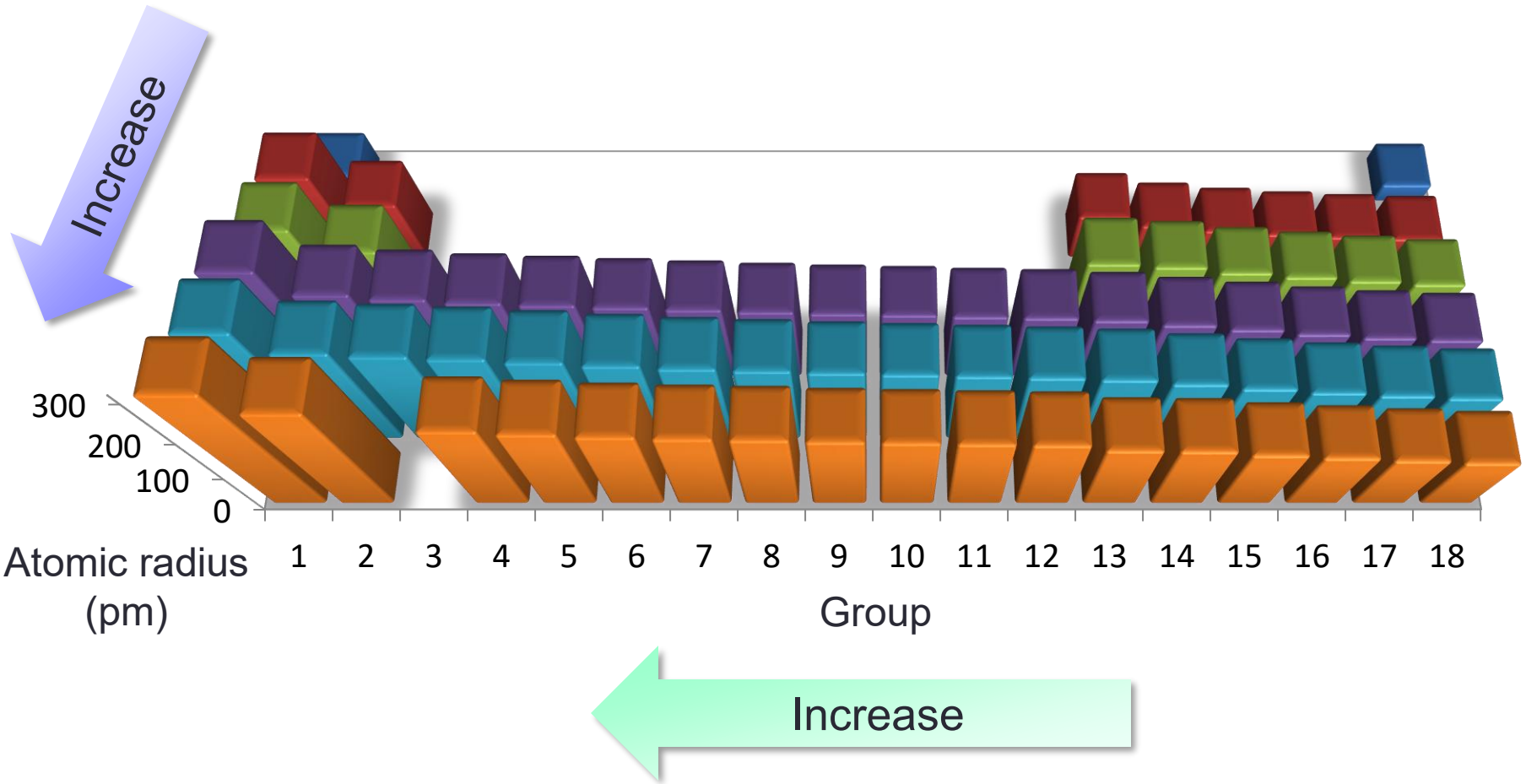
																H	He
																Li	Be
												B	C	N	O	F	Ne
												Al	Si	P	S	Cl	Ar
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn			Ga	Ge	As	Se	Br	Kr
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd			In	Sn	Sb	Te	I	Xe
Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg			Tl	Pb	Bi	Po	At	Rn
Lr																	





1964

Periodic Trends in Atomic Radii

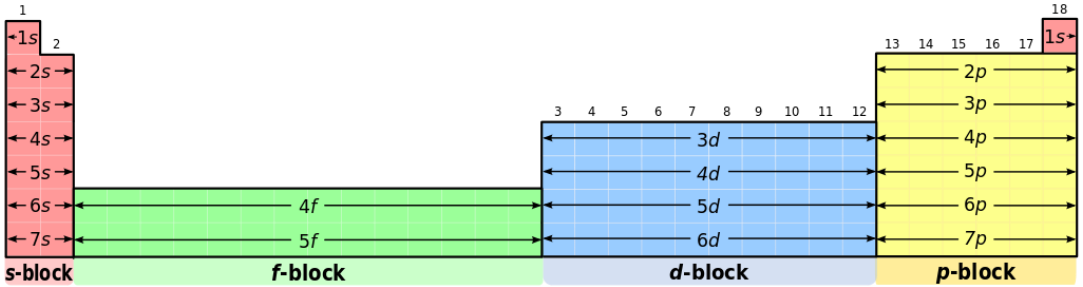


Summary

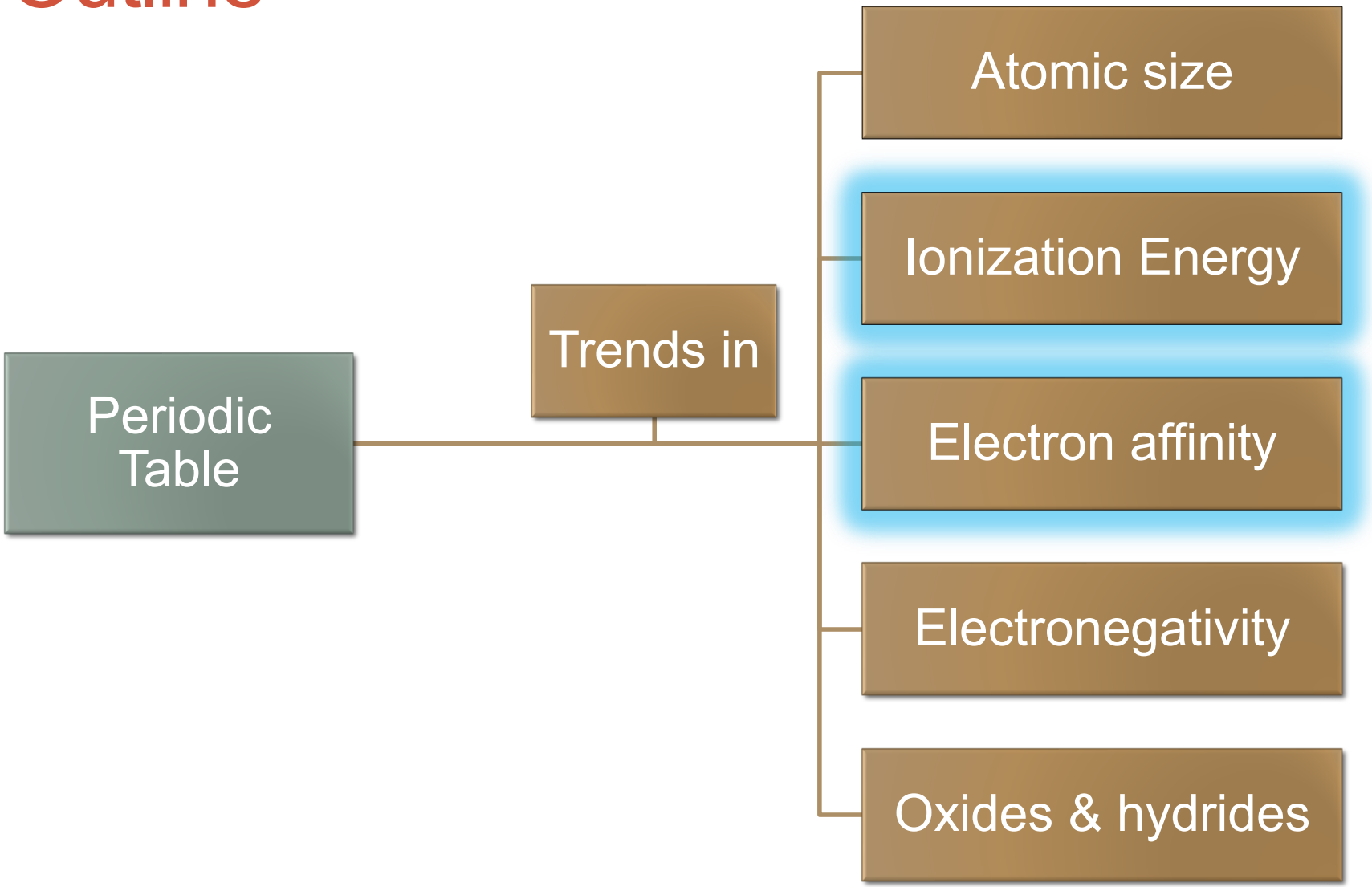
- Atomic radius is determined by the valence *s* subshell
- When estimating Z_{eff} for *ns*:
  - *np* electrons only partly screens *ns*;
  - $(n-1)d$ electrons efficiently screens *ns*;
  - $(n-2)f$ electrons completely screens *ns*.

$$r_{n,Z} \approx \frac{n^2}{Z_{eff}} a_0$$

- The trends in atomic radius $r_{n,Z}$
  - Periods according to *n*
  - $r_{n,Z}$ decreases within the same period
  - *s*-, *p*-, and *d*-blocks
- The Periodic Table



Outline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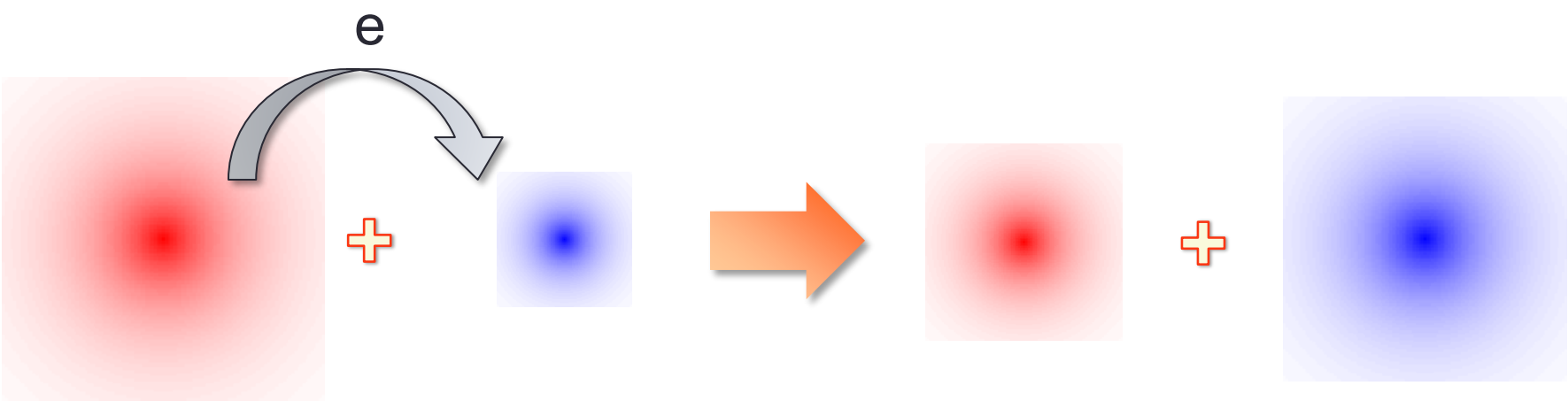
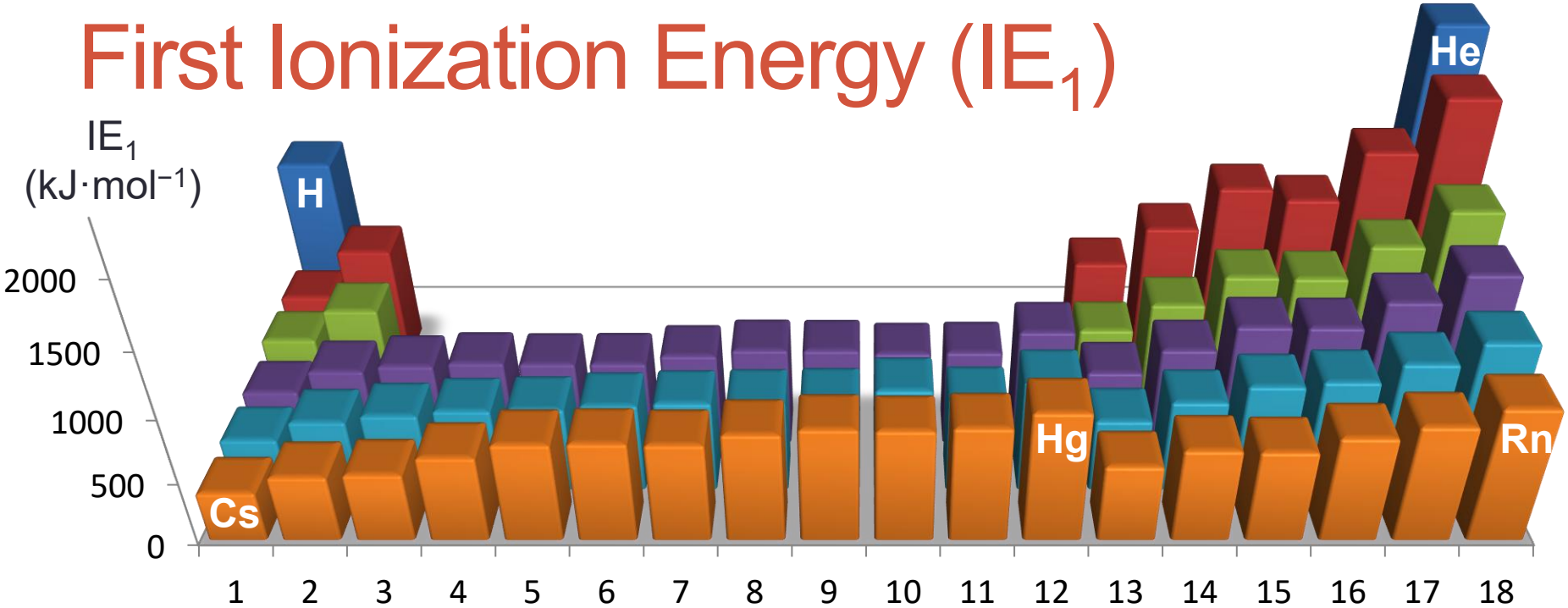
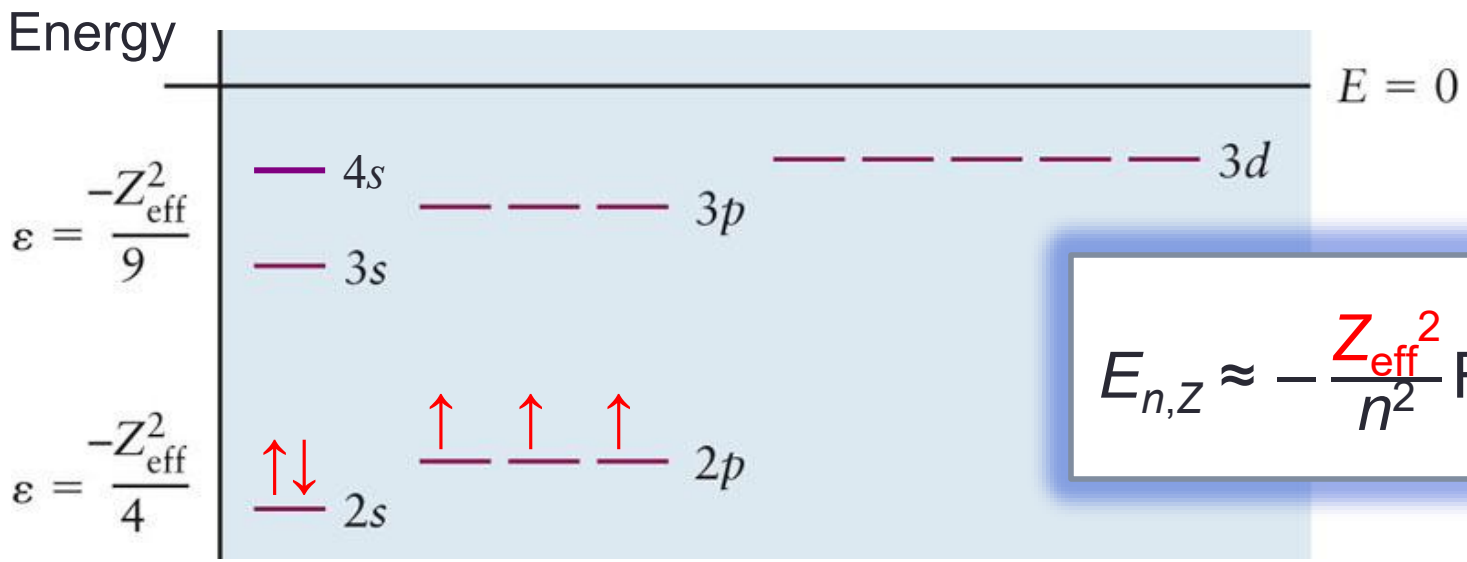
Gaining and Losing an Electron



Na	Cl	Na^+	Cl^-
r = 190 pm	r = 79 pm	r = 116 pm	r = 167 pm

First Ionization Energy (IE_1)





$$E_{n,Z} \approx -\frac{Z_{eff}^2}{n^2}\,\mathrm{Ry}$$

Anomaly of the Trend in IE

Relativistic effect of 6*s* electrons

$$v \rightarrow c \;\Rightarrow\; m_e' = \frac{m_e}{\sqrt{1-v^2/c^2}} \uparrow$$

$$\Rightarrow\; \mathrm{Ry} = \frac{e^4 m_e}{8\varepsilon_0{}^2 h^2} \uparrow \;\Rightarrow\; E_{n,Z} \approx -\frac{Z_{\mathrm{eff}}{}^2}{n^2}\mathrm{Ry} \downarrow$$

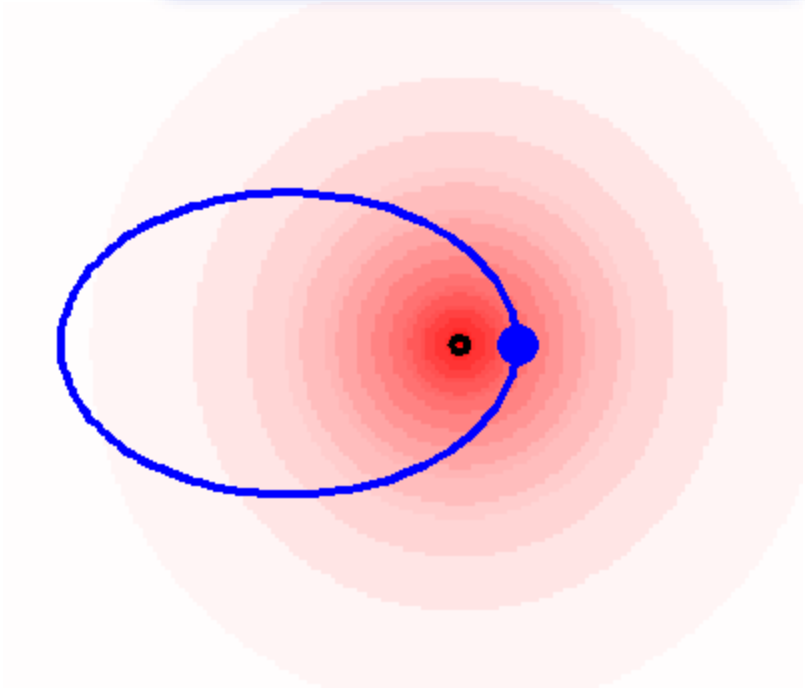
For $_{80}$Hg $6s^2$,

$v/c \approx 80/137 \approx 0.58$

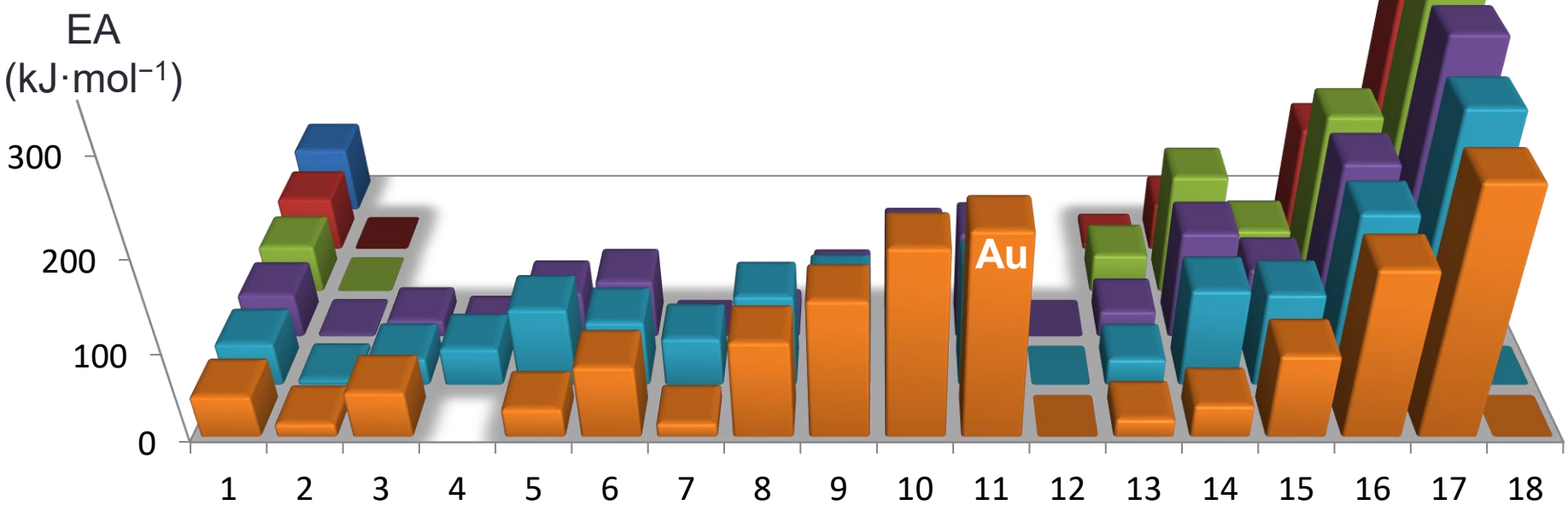
$m_e' = 1.23 m_e$

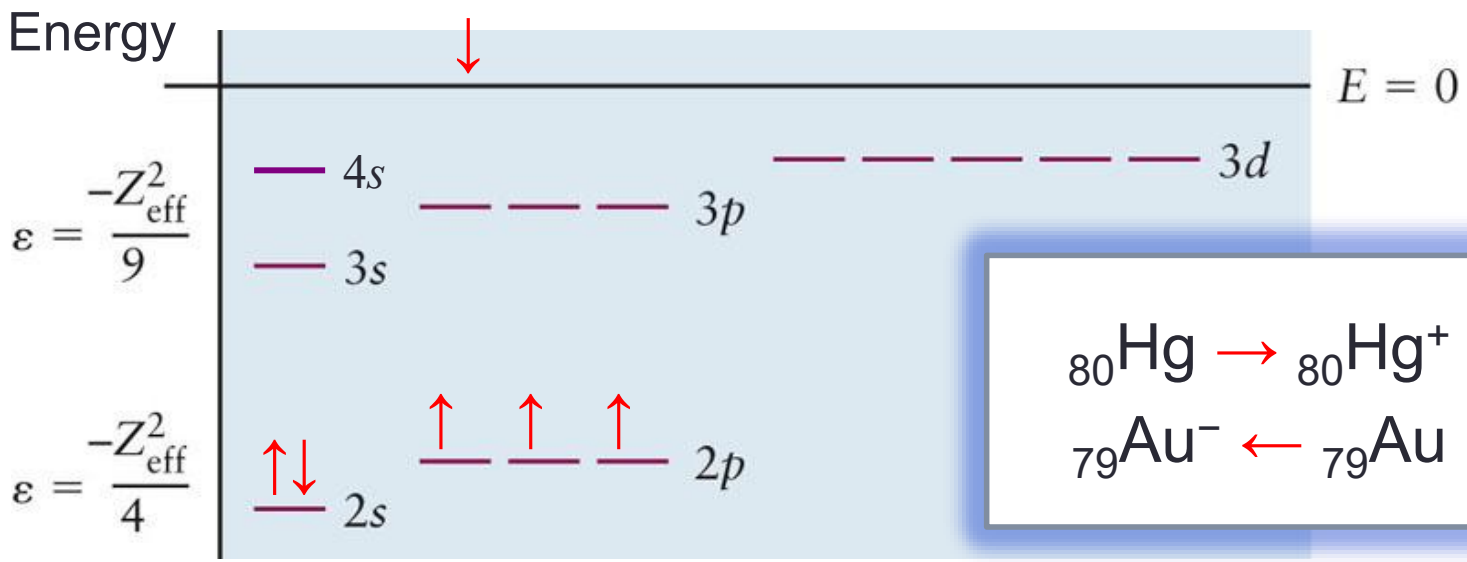
Electron configurations

- Hg: [Xe] $4f^{14}\ 5d^{10}\ 6s^2$
- Hg^+: [Xe] $4f^{14}\ 5d^{10}\ 6s^1$
- Tl: [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^1$



Electron Affinity 电子亲和能

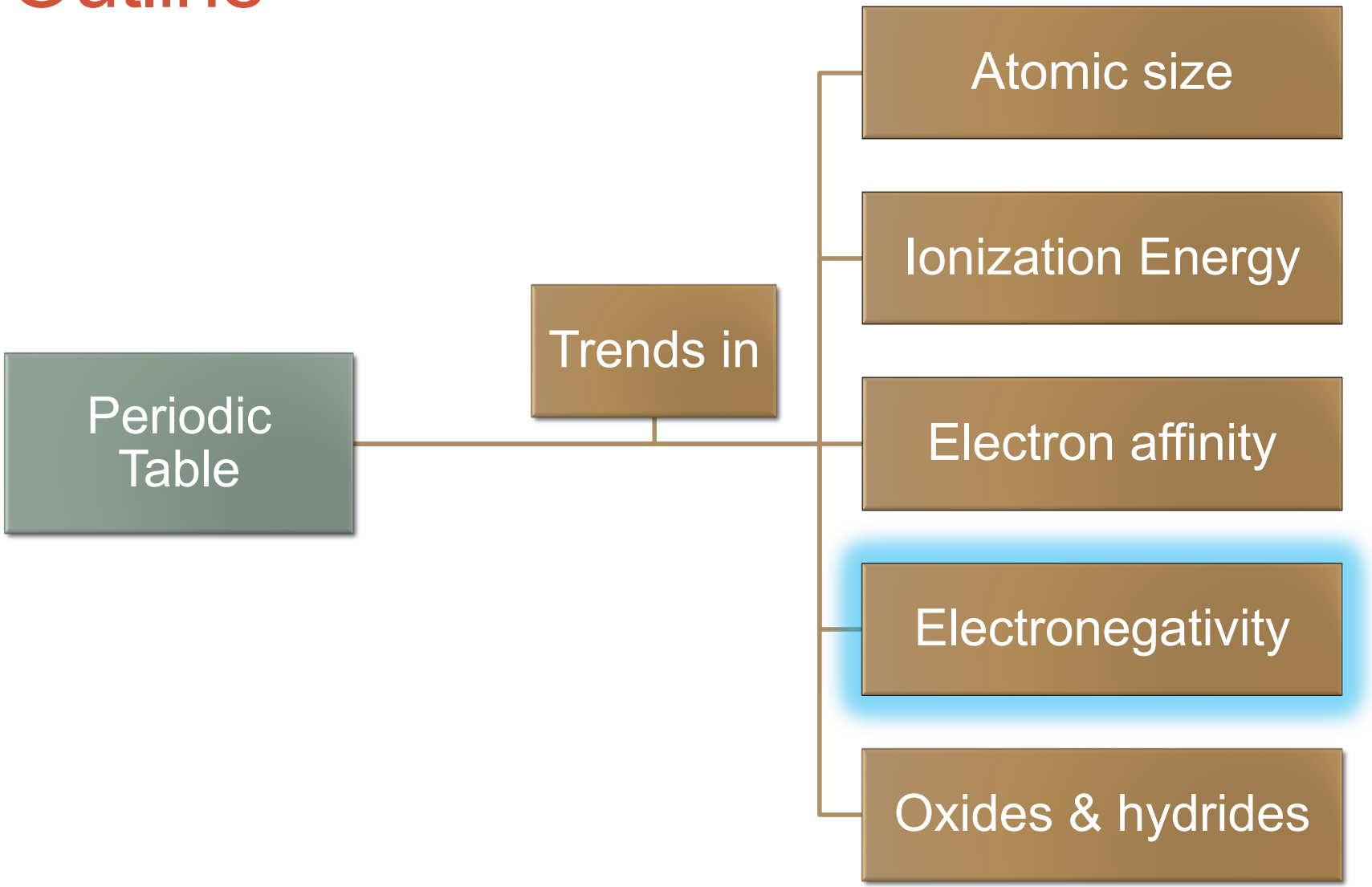




$_{80}Hg \rightarrow {}_{80}Hg^{+} + e$

$_{79}Au^{-} \leftarrow {}_{79}Au + e$

Outline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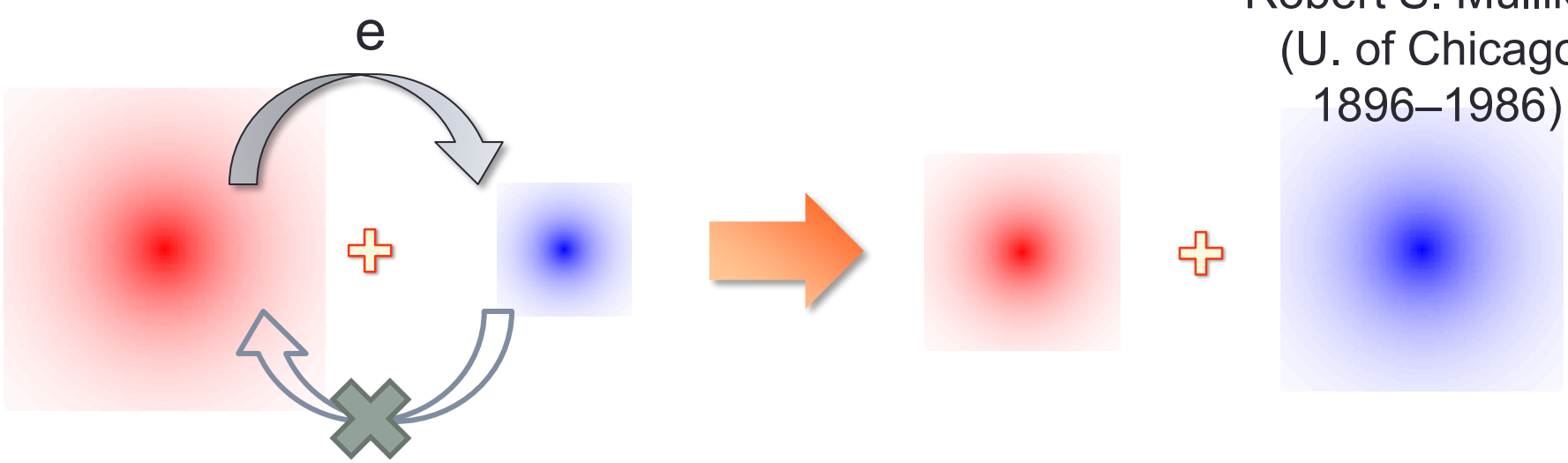
Mulliken's Electronegativity (1934)
电负性

Tendency to attract electrons from a nearby atom.

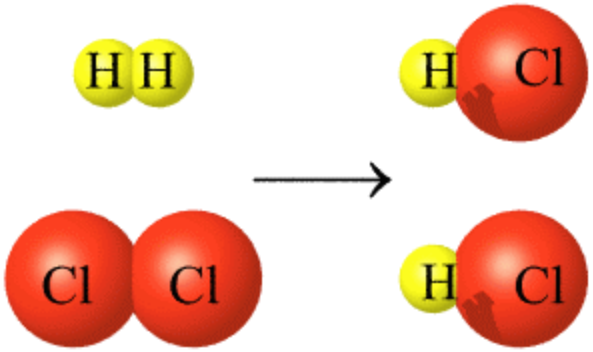
$$\chi \propto \frac{1}{2}(IE_1 + EA)$$

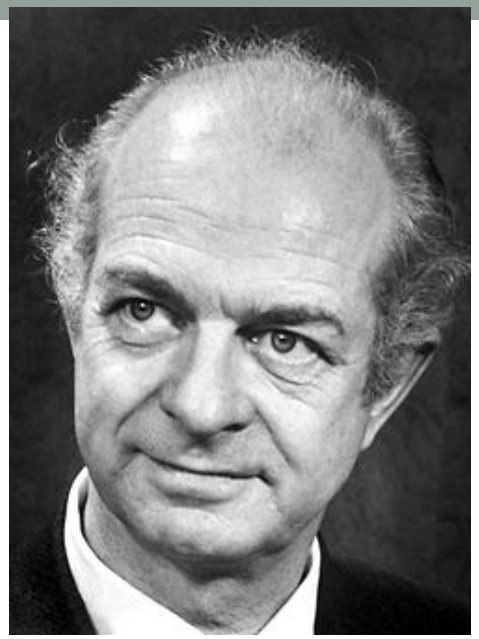
Robert S. Mulliken
(U. of Chicago,
1896–1986)



Na	Cl	Na^+	Cl^-
r = 190 pm	r = 79 pm	r = 116 pm	r = 167 pm

Pauling's Electronegativity (1932)





Linus Pauling
(Caltech, 1901–1994)

$H_2 + Cl_2 \rightarrow 2\ HCl$

$\Delta E = -95$ kJ/mol = -0.98 eV/molecule

$|x_{Cl} - x_H| \approx \sqrt{0.98} = 0.99$

$H_2 + Br_2 \rightarrow 2\ HBr$

$\Delta E = -53$ kJ/mol = -0.55 eV/molecule

$|x_{Br} - x_H| \approx \sqrt{0.55} = 0.74$

$Br_2 + Cl_2 \overset{?}{\rightarrow} 2\ BrCl$

$\Delta E = -1.0$ kJ/mol = -0.01 eV/molecule

$|x_{Cl} - x_{Br}| \approx \sqrt{0.01} = 0.1$

Pauling's Electronegativity

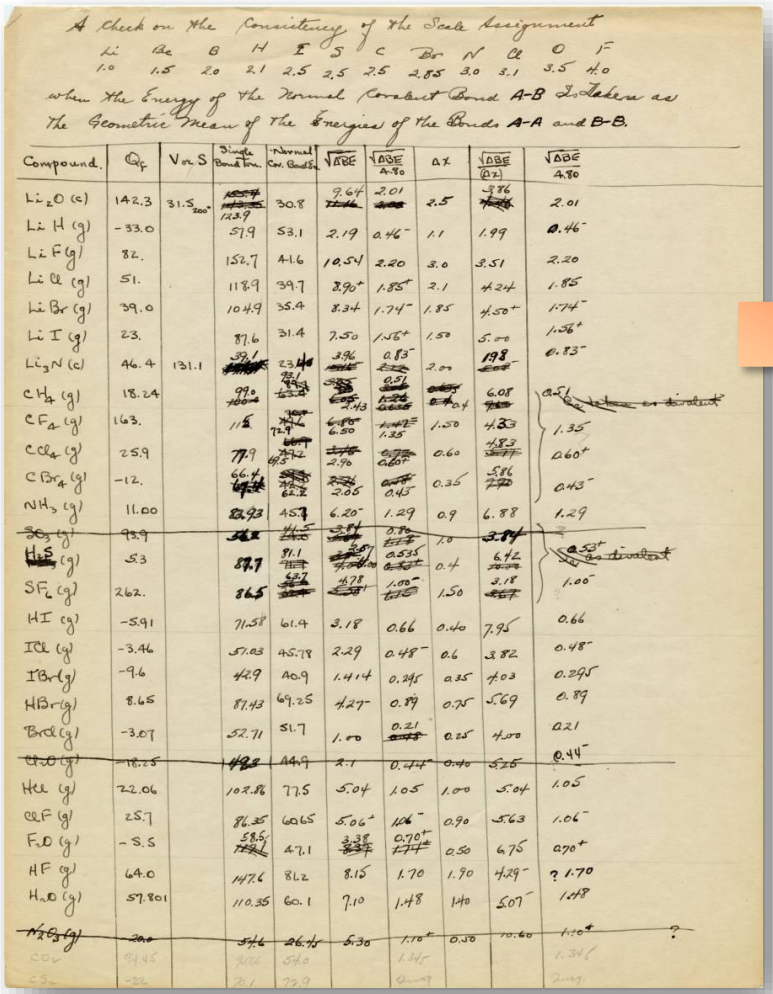
$$D_{AB} - {}^1/_2(D_{AA} + D_{BB}) = k(\Delta\chi)^2$$

$$D_{AB} - (D_{AA}D_{BB})^{1/2} = k(\Delta\chi)^2$$

$$D_{AB} - {}^1/_2(D_{AA} + D_{BB}) = k|\Delta\chi|$$

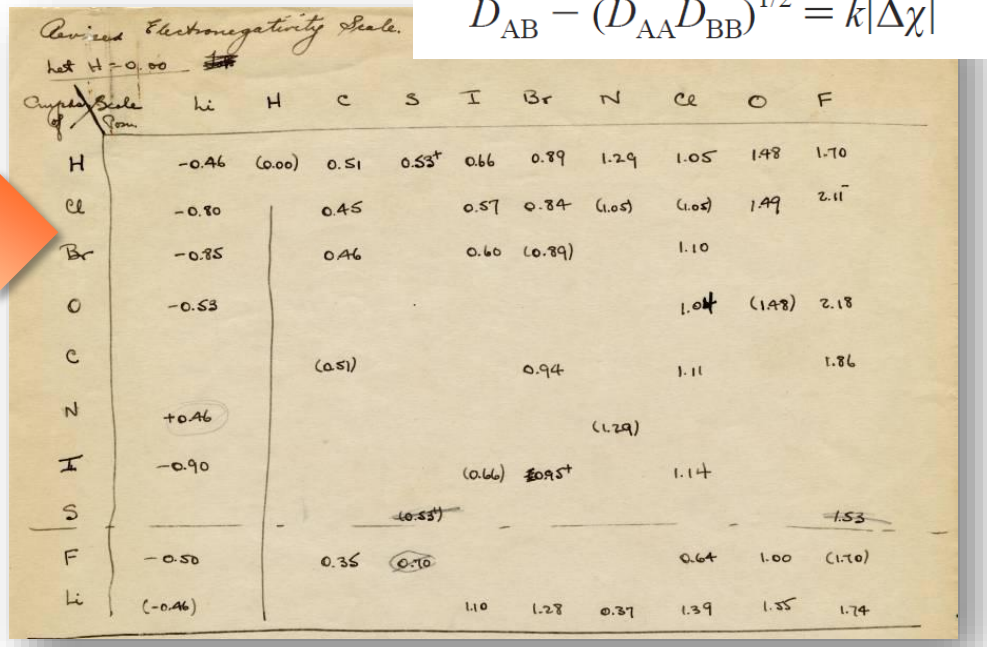
$$D_{AB} - (D_{AA}D_{BB})^{1/2} = k|\Delta\chi|$$

Experimental data D_{AA}, D_{BB}, D_{AB}



Fitting gives

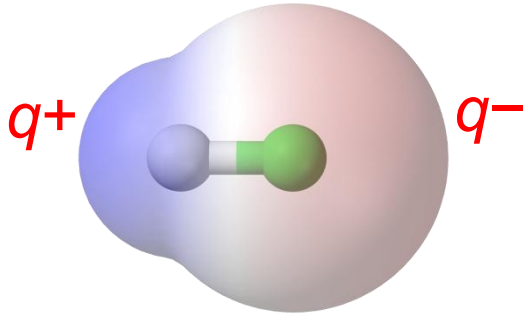
$\Delta\chi = \chi_A - \chi_B$



Individual χ_A and χ_B
on a scale between 0 and 4

https://scarc.library.oregonstate.edu/coll/pauling/bond/

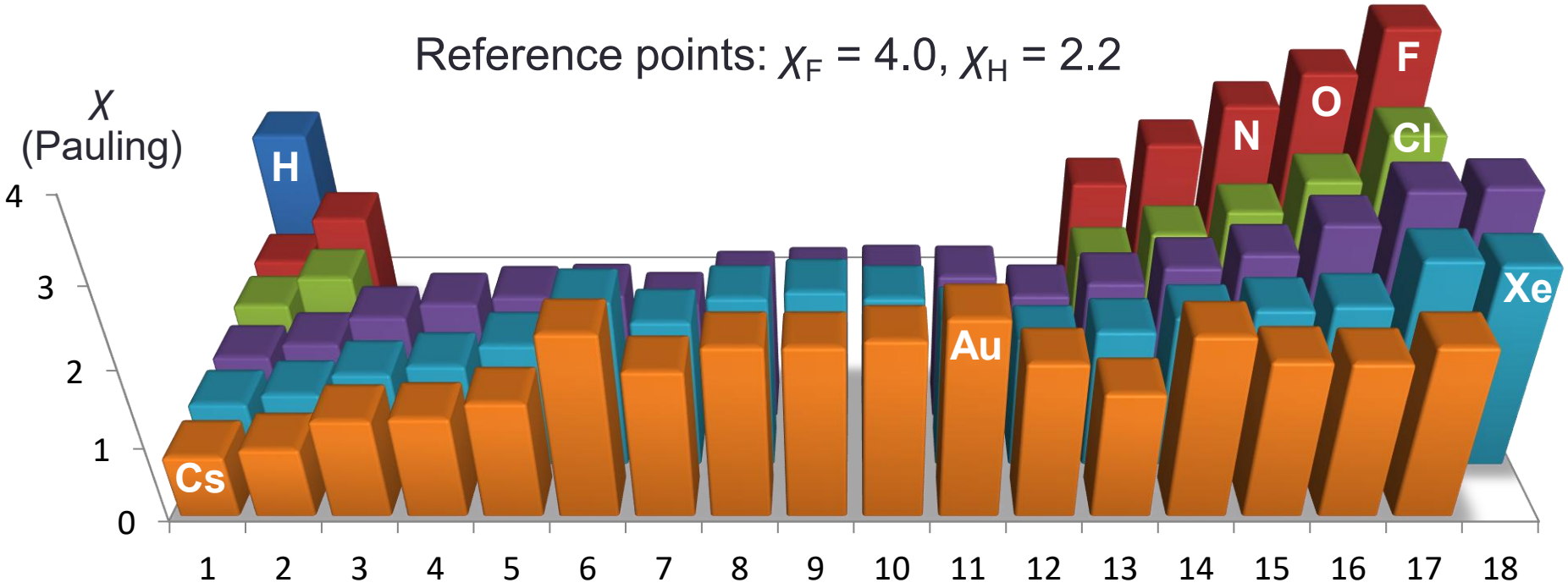
Calculating Electronegativities



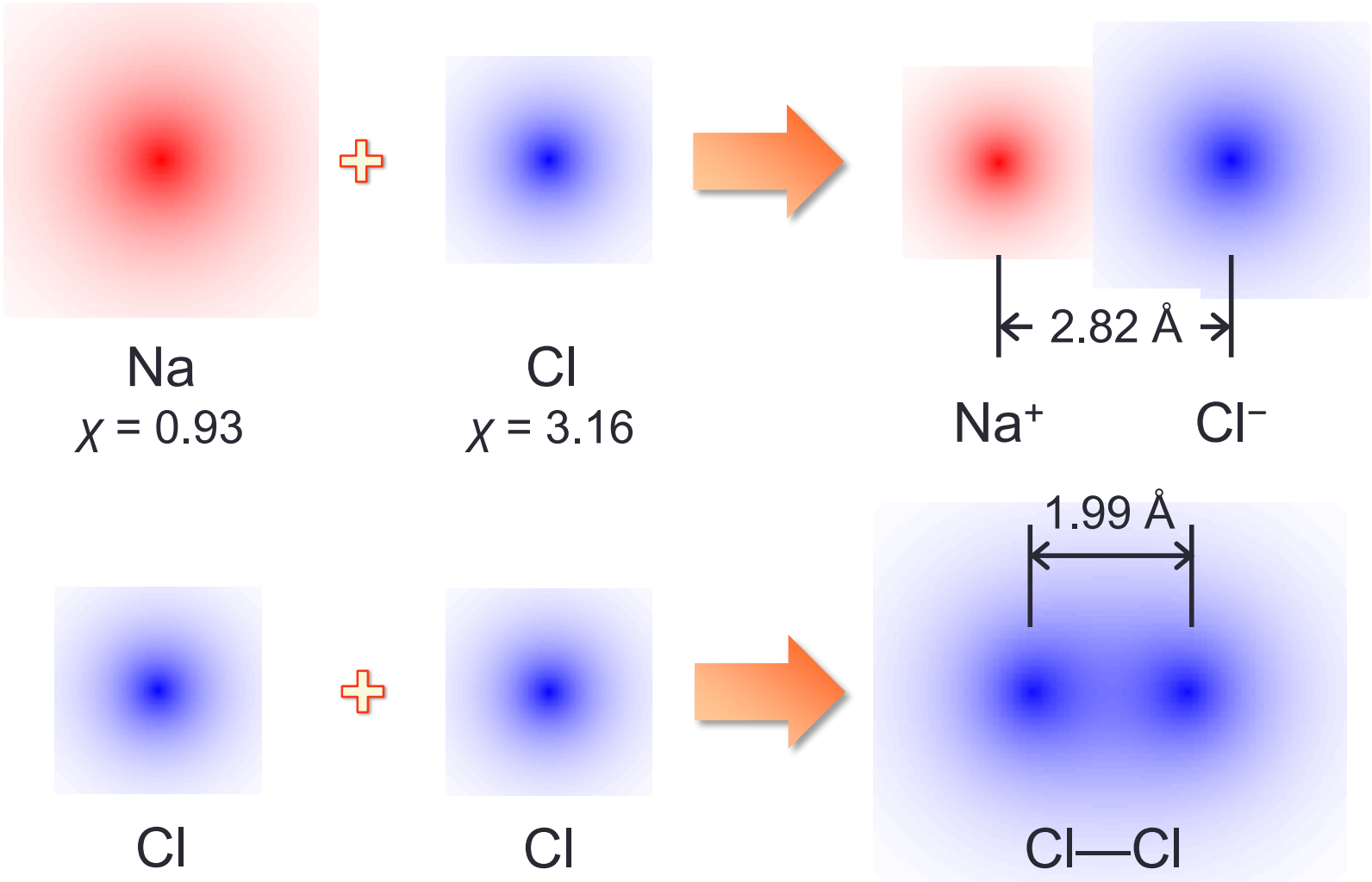
H^{q+}—Cl^{q-}

$$(\chi_A - \chi_B)^2 \propto q^2 \propto \Delta E$$

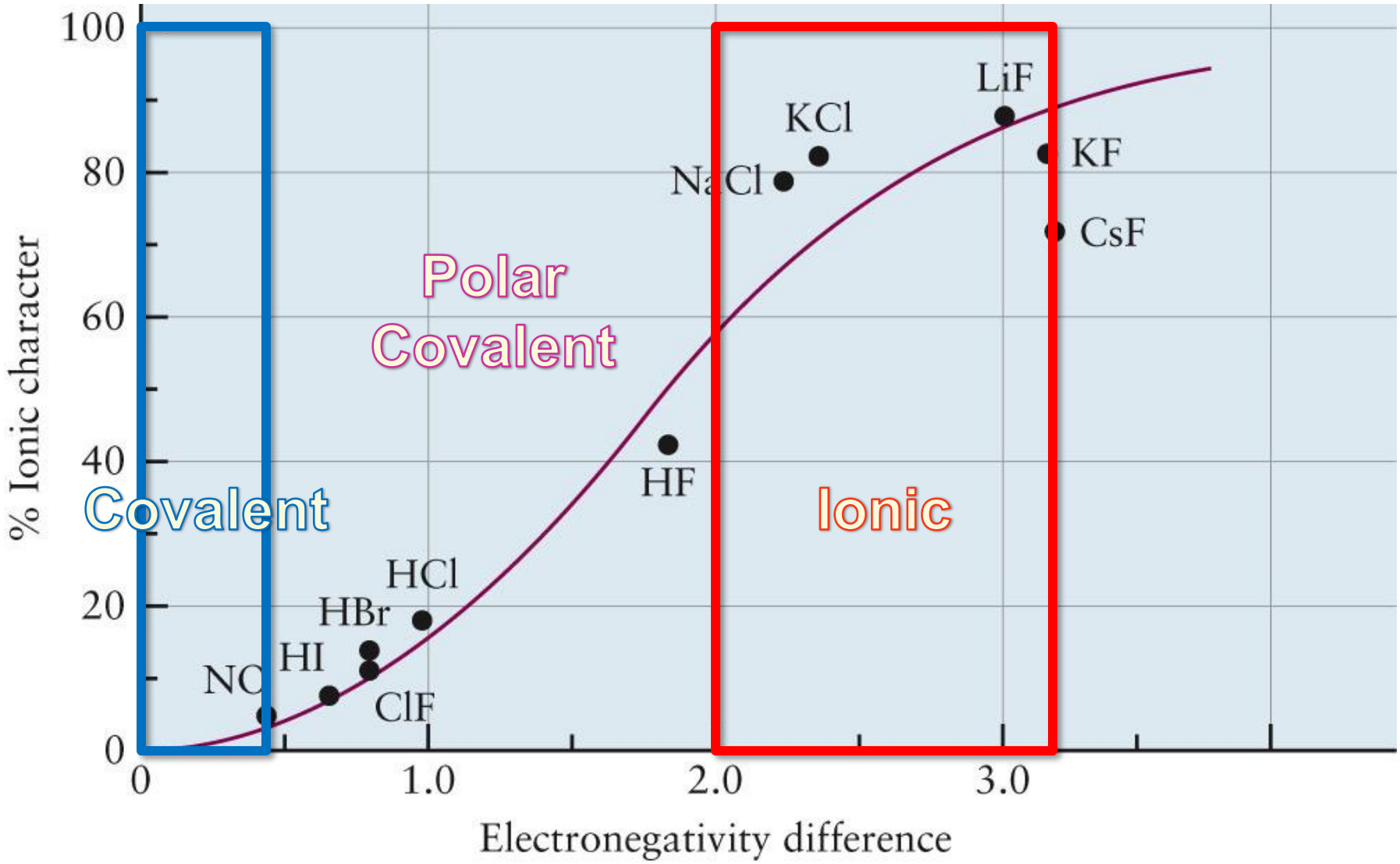
Reference points: $\chi_F = 4.0$, $\chi_H = 2.2$



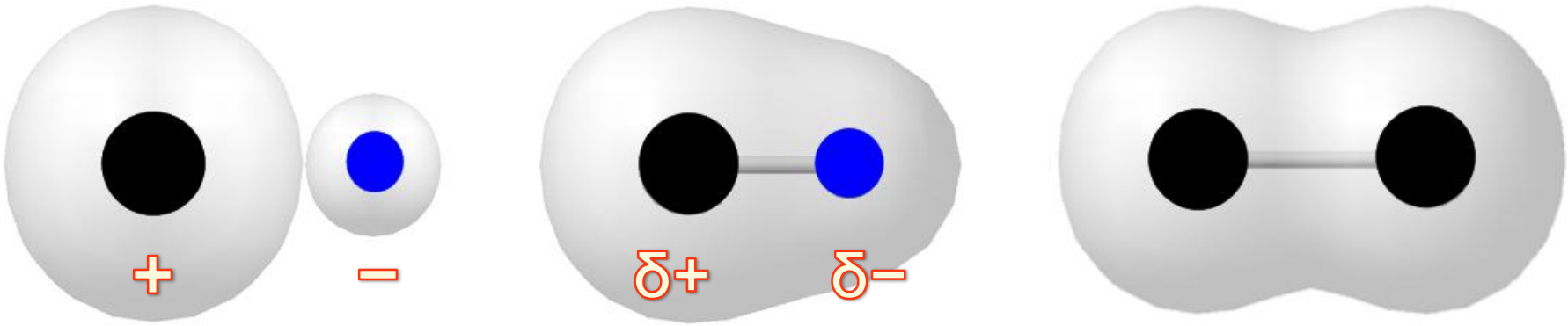
Chemical Bonds



Electronegativity and Bonding



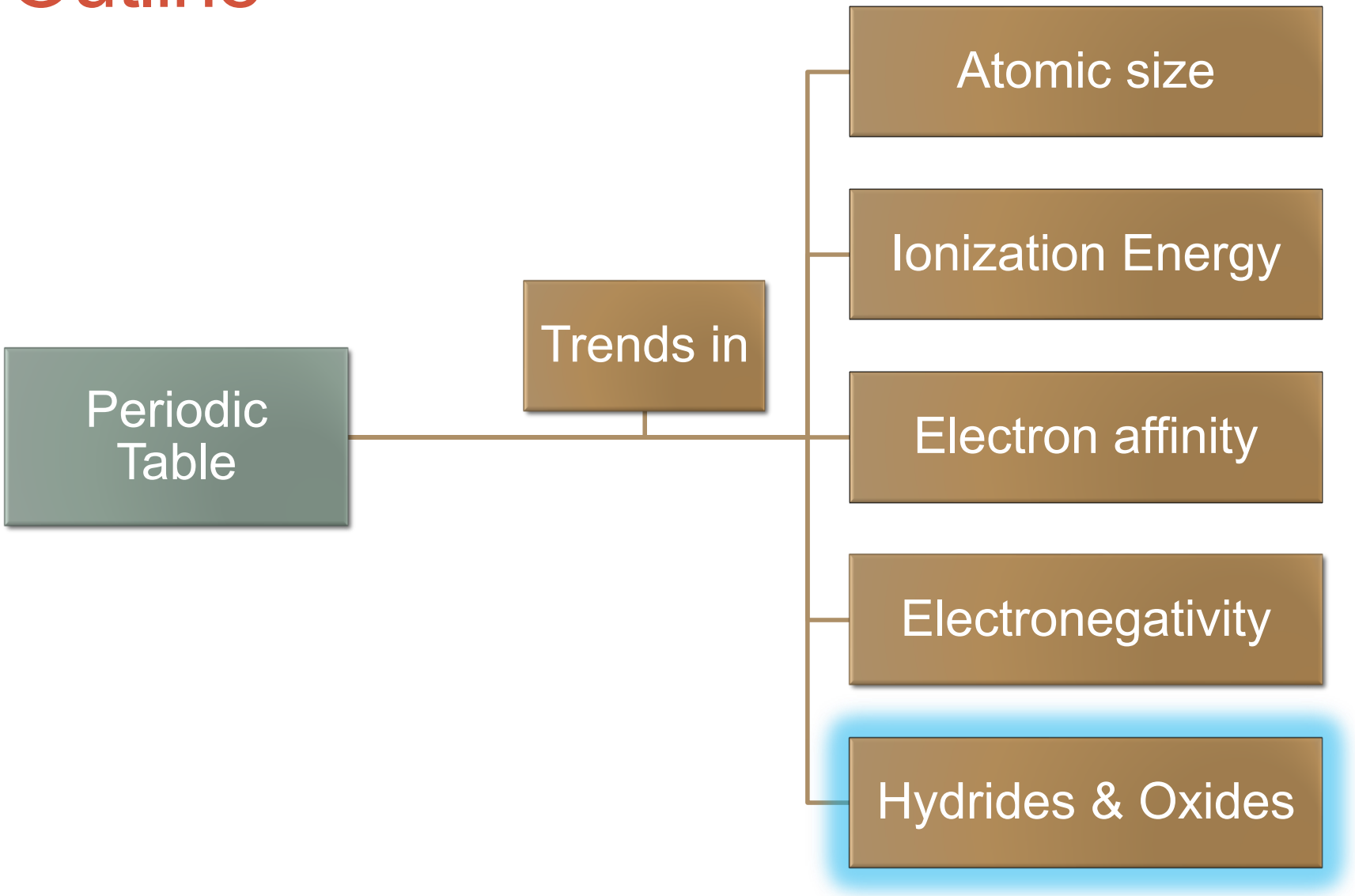
From Ionic to Covalent Bonding



Ionic	Polar Covalent	Covalent
$\Delta\chi > 2.0$		$\Delta\chi < 0.4$
Low positive charge		High positive charge
Large cation		Large anion

Higher charge ⇒ Higher χ

Outline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Hydrides & Oxides

Periodic Trends in Forming Hydrides

LiH	BeH_2	BH_3	CH_4	NH_3	H_2O	HF
NaH	MgH_2	AlH_3	SiH_4	PH_3	H_2S	HCl

Ionic ← Covalent → Polar

χ = 2.20

χ = 2.55

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	**															

Periodic Trends in Forming Oxides

+5 δ− δ+

Li_2O BeO B_2O_3 CO_2 N_2O_5 +6 OF_2

Na_2O MgO Al_2O_3 SiO_2 P_2O_5 SO_3 +7 Cl_2O_7

Ionic ⟷ Covalent

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	**															

Acid/Base Properties of Oxides

Basic:

$$Na_2O + H_2O \rightarrow Na^+(aq) + OH^-(aq)$$

Acidic:

$$Cl_2O_7 + H_2O \rightarrow ClO_4^-(aq) + H^+(aq)$$

Amphoteric 两性的:

$$Al_2O_3 + H^+(aq) \rightarrow Al^{3+}(aq) + H_2O$$

$$Al_2O_3 + OH^-(aq) + H_2O \rightarrow Al(OH)_4^-(aq)$$

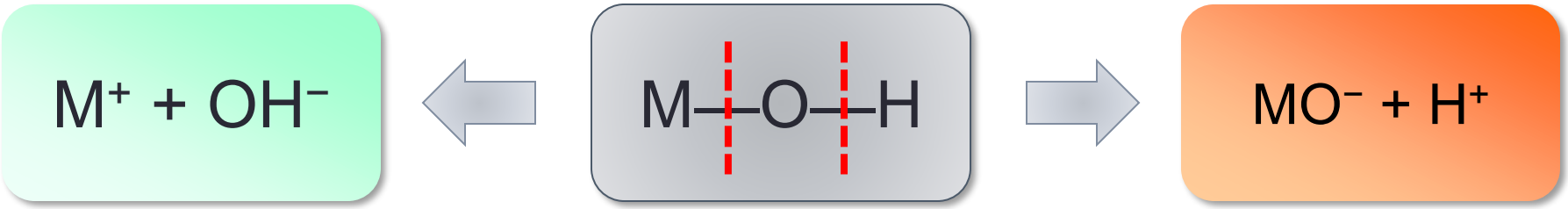
aq: aqueous 水合

Most acidic: $HClO_4$
Most basic: CsOH

Acidic

H																	He
Li	Be	*Basic*										B	C	N	O	F	Ne
Na	Mg						*Amphoteric*					Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn

Electronegativity and Bond Breaking



$$\chi_O = 3.5$$
$$\chi_H = 2.2$$
$$\chi_M = ?$$

Summary

- Electronegativity
  - Mulliken: $\chi \propto \frac{1}{2}(IE_1 + EA)$
  - Pauling: $(\chi_A - \chi_B)^2 \propto q^2 \propto \Delta$
- Bonding
  - Ionic
  - Covalent / polar covalent
- Oxides and hydrides
  - Charge
  - Basic, acid, amphoteric

$\chi_F = 4.0$
$\chi_H = 2.2$

Next Lecture Series: Bonding in Molecules

Reading: OGB8 §6, YY §§3.2–3.5